Effective Archimedean Stark Theorems over Real Quadratic Fields:
A Uniform Quadratic Formula, Higher-Order Packets, and CM Descent

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Abstract

We prove a uniform formula for every one-place real-quadratic Stark invariant whose differenced Fourier support is quadratic. The formula is an explicit product of relative units, with rational exponents computed from class numbers, imprimitive Euler factors, and an exact relative-regulator index. Beyond this quadratic floor, we identify seven packets by Shintani's index-two theorem and effective height rigidity, including support orders six and ten, and prove further quartic packets by descent to Stark's proved imaginary-quadratic rank-one theorem. One packet is obtained independently from Shintani's 1978 theorem and Stark's 1980 theorem. A difficult $\mathbb{Q}(\sqrt{6})$ example is closed through two auxiliary primes and two CM bases with $e = 8$ and $e = 12$. Two structural lemmas show that a nontrivial one-place packet cannot be governed by an absolutely abelian ray field and must have even Shintani index. The latter is independently consistent with all 446 genuinely reconstructed odd-index cases in a separate census. The paper contains the mathematical proofs; exact and Arb certificates provide independently replayable checks. No unproved Stark conjecture or floating-point recognition enters a proof.

1 Introduction and claim boundary

Let K be real quadratic and let $\mathfrak{m} = \mathfrak{f}\infty_2$ be a ray modulus containing one real place. For $A \in \mathrm{Cl}_{\mathfrak{m}}(K)$, let R be the sign class and put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \qquad X_A = \exp Z'_{\mathfrak{m}}(0, A). \tag{1}$$

The numbers X_A are positive real analytic invariants. We call the finite Artin-indexed tuple

$$\mathcal{X}_{\mathfrak{m}} = (X_A)_{A\in \mathrm{Cl}_{\mathfrak{m}}(K)}$$

the *one-place packet*; repeated values may be suppressed when a smaller Artin orbit is displayed. The word "packet" carries no algebraicity assumption. Stark's rank-one conjecture predicts algebraic units with the prescribed Artin action, but that conjecture is not known in this generality. The purpose of this paper is to identify explicit packets only where a proved theorem already supplies algebraicity and where the remaining analytic-to-algebraic comparison can be certified.

The word *unconditional* has this precise meaning throughout: each displayed identity follows from a published proved theorem under hypotheses checked exactly in the case bundle. It does not mean that the general real-quadratic rank-one Stark conjecture is proved. Shintani's theorem is used only in its index-two range [1]; Stark's theorem is used only for abelian extensions of imaginary quadratic fields [2]; and the quadratic cases use the analytic class-number formula. Numerical PARI recognition, when retained as a cross-check, is outside the proof chain.

Terminology and quantitative conventions

Fourier inversion of (1) involves precisely the ray characters χ for which $\chi(R) = -1$. The *support order* is the maximum order of such a character. Labels such as `RQ-000190` are stable locators in the accompanying corpus; they are not mathematical notation, and every use below also states the field and modulus.

For Engine B, a *safe exponent* is an explicitly computed integer m for which Shintani's theorem proves that every X_A^m is algebraic in the asserted ray field. If the analytic and algebraic logarithms differ by balls of radius at most δ, the height comparison bounds the quotient by $m\delta$. The reported *margin* is

$$\mathcal{M} = \frac{V_{\min}}{m\delta}, \tag{2}$$

where $V_{\min}$ is the minimum applicable Voutier lower bound over the certified degree window. Thus $\mathcal{M} > 100$ is the pre-registered acceptance condition. A *cone evaluator* is the finite Shintani–Yamamoto decomposition of a partial zeta derivative into double-sine or Mellin-integral terms; all such terms below are evaluated as Arb balls.

How the results fit together

There are three theorem mechanisms, not a sequence of unrelated case studies. Engine A is a uniform closed-form floor for quadratic Fourier support. Engine B crosses to higher character orders when Shintani's maximal absolutely abelian subfield has index two; the order-six cases on either side of ramification isolate this index/wildness condition as the relevant obstruction. Engine C handles quartic support through imaginary-quadratic descent and gives a theorem base disjoint from Engine B. The dual-routed case tests their overlap. Finally, Lemmas 4 and 5 show why neither an absolutely abelian fourth engine nor character order alone describes the frontier.

Main theorems and epistemic status

Theorem 1 (Uniform quadratic-support theorem)**.** *For every real quadratic K, every one-place modulus $\mathfrak{m} = \mathfrak{f}\infty_2$, and every packet whose differenced Fourier support consists only of quadratic characters, each X_A is an explicit rational power of a product of relative units. Formula* (5) *gives the exponents entirely in terms of exact class numbers, roots of unity, imprimitive Euler factors, and relative-regulator indices. No numerical approximation is required to state or prove the formula.*

Theorem 2 (Selected higher-order and CM packets)**.** *The following one-place archimedean packets are identified unconditionally by Engine B.*

case	K	$\mathrm{N}\mathfrak{f}$	*support*	*safe exponent*	*margin*
RQ-000190	$\mathbb{Q}(\sqrt{7})$	*7*	$2, 6$	*4032*	> 5688
RQ-000419	$\mathbb{Q}(\sqrt{14})$	*7*	$2, 6$	*4032*	> 7315
RQ-000108	$\mathbb{Q}(\sqrt{5})$	*45*	4	*2880*	> 2460
RQ-000021	$\mathbb{Q}(\sqrt{2})$	*49*	$2, 6$	*2016*	> 4261
RQ-002057	$\mathbb{Q}(\sqrt{57})$	*27*	$2, 6$	*2592*	> 748
RQ-002955	$\mathbb{Q}(\sqrt{77})$	*7*	$2, 6$	*4032*	> 5151
RQ-001107	$\mathbb{Q}(\sqrt{33})$	*11*	$2, 10$	*15840*	> 5817

Here "margin" has the meaning in (2). In addition:

1. *the primitive norm-32 packet over $\mathbb{Q}(\sqrt{35})$, and its norm-64 transport, are proved independently through $\mathbb{Q}(\sqrt{-10})$ and $\mathbb{Q}(\sqrt{-14})$;*
2. *three generic CM-descent packets, RQ-001569, RQ-007519, and RQ-001894, are proved independently through two imaginary quadratic bases each, with $e = |\mu(E)| = 6$;*
3. *the norm-8 packet RQ-000129 over $\mathbb{Q}(\sqrt{6})$ is proved through the $e = 8$ and $e = 12$ CM routes after an exact auxiliary-prime enlargement; and*
4. *the quartic packet RQ-000458 over $\mathbb{Q}(\sqrt{14})$ is proved once by Shintani's theorem and once by Stark's imaginary-quadratic theorem.*

Epistemic ledger. Theorems 1 and 2, and every item enumerated in Theorem 2, have mathematical status `VERIFIED`. RQ-000458 has the process label `DUAL_ROUTED`, rather than `DUAL_PROVED`, because its second-route selection seal was not timestamped before execution; both mathematical proofs remain valid. A dated literature search found no earlier unconditional oriented examples with support order six or ten, so these are apparently the first examples in those order classes. This historical observation is not part of either theorem and is not asserted as a universal priority claim.[1]

Theorem inventory

For clarity, the paper's contributions are the following eight items. The status in this list concerns mathematical proof unless it is explicitly prefixed by "historical" or "process."

1. **Order six and its replication (`VERIFIED`).** The packet over $\mathbb{Q}(\sqrt{7})$ has margin > 5688, and the structural replication over $\mathbb{Q}(\sqrt{14})$ has margin > 7315. The dated literature search records them as apparently the first unconditional oriented packets with support order six (`SURVEYED_PRIORITY`, not a theorem claim). Together with the separate $\mathbb{Q}(\sqrt{21})$ failure, they show that character order six is not itself the obstruction; the operative boundary is the index-two/wildness hypothesis.
2. **Order ten (`VERIFIED`).** The $\mathbb{Q}(\sqrt{33})$ packet has margin > 5817 and is the first proved support order in the present inventory containing the prime five. Its apparent historical priority has status `SURVEYED_PRIORITY`.
3. **Ramified-prime-3 control (`VERIFIED`).** RQ-002057 over $\mathbb{Q}(\sqrt{57})$, with finite modulus of norm 27, has order-six support and wild 3-power conductor structure. It is the closest proved structural neighbor of the $\mathbb{Q}(\sqrt{21})$ wall and sharpens the diagnosis in item 1.
4. **Uniform Engine-A theorem (`VERIFIED_THEOREM`).** Theorem 1 gives a closed product formula for every quadratic-support packet, reducing each application to exact class numbers, Euler factors, unit kernels, and a relative-regulator index. This is the paper's broadest result and is quotable independently of the corpus or its software.
5. **Two disjoint theorem routes (`VERIFIED`; process `DUAL_ROUTED`).** RQ-000458 is the same Artin-labeled packet proved once through Shintani 1978 and once through Stark 1980, with disjoint theorem bases and intermediate data. The deliberately weaker process label records the missing contemporaneous selection seal; it does not weaken either proof.
6. **Generic CM closure beyond class number one (`VERIFIED`).** The $\mathbb{Q}(\sqrt{35})$ packet is the first generic Engine-C closure in the corpus with $h_K > 1$. Two imaginary bases of different class numbers, 2 and 4, are forced to the same Artin-labeled packet.

[1] The search record names the sources, versions, hashes, and date. Kopp's arXiv:2411.06763 was checked again on 30 July 2026; the current version remains v3, dated 3 May 2025, and its algebraicity applications are conditional.

7. **General-e CM closure (`VERIFIED`).** The $\mathbb{Q}(\sqrt{6})$ packet is proved through $e = 8$ and $e = 12$ routes after auxiliary-prime enlargement. Independent primes verify that the enlargement is harmless, while the exact identity $q_8^3 = q_{12}^2$ aligns the two values of e.

8. **No-go lemma (`VERIFIED_THEOREM`).** Lemma 4 proves that no nontrivial one-place differenced packet over a real quadratic field is governed by an absolutely abelian ray field. It rules out a hidden fourth abelian engine and therefore proves a small completeness statement about the method taxonomy itself.

Theorem 2 is deliberately a selected-results theorem, not a census-completeness statement. Counts, conductor trends, and a complete frontier taxonomy require the separately audited census and will appear elsewhere. This separation prevents provisional classification data from entering a theorem proof. The prior numerical and conditional computations of Dummit–Sands–Tangedal and Cohen–Roblot remain prior work rather than claims displaced here [6, 7, 8, 9]. More precisely, the 1997 Dummit–Sands–Tangedal computations concern totally real cubic bases and are numerical; their 2003 exponent-two theorems overlap the theorem class of Engine A, which is consequently not advertised as a historical first. Cohen–Roblot construct Hilbert class fields conditionally from Stark units and verify the resulting fields independently, whereas every promoted object here is a one-place ray field with nontrivial finite modulus. Thus these are mathematically adjacent precedents, not identical class-field objects. The order-six and order-ten historical statements refer only to the dated search described above.

2 Three proved engines

2.1 Engine A: quadratic characters

Let $G = \mathrm{Cl}_{\mathfrak{m}}(K)$, and suppose every character χ with $\chi(R) = -1$ has order two. Write χ° for the associated primitive Hecke character and L_χ/K for the quadratic extension it cuts out. The selected real place becomes complex in L_χ, while the other real place splits, so L_χ has signature $(2, 1)$.

Let ϵ_K generate $E_K/\mu(K)$. Modulo torsion, use the logarithmic embedding

$$\lambda_L(u) = \big(\log|\tau_1 u|, \log|\tau_2 u|, 2\log|\tau_3 u|\big), \tag{3}$$

where τ_1, τ_2 are real and τ_3 is one of the complex embeddings. Let u_χ generate the primitive rank-one kernel of

$$N_{L_\chi/K} : E_{L_\chi}/\mu(L_\chi) \longrightarrow E_K/\mu(K),$$

and orient it by $\log|\tau_1 u_\chi| > 0$. In any exact basis of $E_{L_\chi}/\mu(L_\chi)$, write v_K and v_χ for the coordinate columns of ϵ_K and u_χ, and define

$$I_\chi = [E_{L_\chi}/\mu(L_\chi) : \langle E_K/\mu(K), u_\chi\rangle] = |\det(v_K, v_\chi)|. \tag{4}$$

Proposition 3 (Explicit quadratic formula)**.** *With E_χ the exact product of imprimitive Euler factors at zero,*

$$L'_{\mathfrak{m}}(0, \chi) = E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{2}{I_\chi} \log|\tau_1 u_\chi|. \tag{5}$$

Moreover

$$Z'_{\mathfrak{m}}(0, A) = \frac{2}{|G|} \sum_{\chi(R)=-1} \chi(A)^{-1} L'_{\mathfrak{m}}(0, \chi). \tag{6}$$

Consequently a common denominator in (5)–(6) gives an explicit power identity for every X_A.

Proof. For a primitive quadratic χ°, $\zeta_{L_\chi}(s) = \zeta_K(s)L(s,\chi^\circ)$. The analytic class-number formula at zero, with the same regulator normalization as (3), is

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1(F)+r_2(F)-1} + O\Big(s^{r_1(F)+r_2(F)}\Big).$$

Since the unit ranks of K and L_χ are one and two, respectively, division gives

$$L'(0,\chi^\circ) = \frac{h_{L_\chi}}{h_K}\frac{w_K}{w_{L_\chi}}\frac{R_{L_\chi}}{R_K}. \tag{7}$$

We next prove the regulator identity without suppressing the mixed signature. Put $a = \log|\tau_1\epsilon_K|$, where τ_1, τ_2 lie over the split real place of K. The product formula gives

$$\lambda_L(\epsilon_K) = (a, a, -2a).$$

For the relative unit u_χ, exact norm one modulo torsion gives

$$\lambda_L(u_\chi) = (b, -b, 0), \qquad b = \log|\tau_1 u_\chi| > 0.$$

The determinant of the first two coordinates is $2|a|b$. Since $R_K = |a|$, the regulator of $\langle E_K, u_\chi\rangle$ is $2R_K \log|\tau_1 u_\chi|$. The same subgroup has index I_χ in the full free unit group, so its regulator is $I_\chi R_{L_\chi}$. Therefore

$$\frac{R_{L_\chi}}{R_K} = \frac{2}{I_\chi}\log|\tau_1 u_\chi|. \tag{8}$$

If $\mathfrak{m}/\mathfrak{f}_{\chi^\circ}$ is the imprimitive part, then

$$L_{\mathfrak{m}}(s,\chi) = L(s,\chi^\circ)\prod_{\mathfrak{p}\mid\mathfrak{m}/\mathfrak{f}_{\chi^\circ}}\left(1-\chi^\circ(\mathfrak{p})N\mathfrak{p}^{-s}\right).$$

At $s=0$ the product is the exact integer E_χ. If one factor vanishes, both this product and $L(s,\chi^\circ)$ vanish at zero, so $L'_{\mathfrak{m}}(0,\chi) = 0$; otherwise combining (7) and (8) proves (5).

Finally, character inversion gives

$$\zeta_{\mathfrak{m}}(s,A) = \frac{1}{|G|}\sum_{\chi\in\widehat{G}}\chi(A)^{-1}L_{\mathfrak{m}}(s,\chi).$$

Subtracting the formula with A replaced by RA multiplies the χ-term by $1-\chi(R)^{-1}$, which is two precisely for $\chi(R) = -1$ and zero otherwise. Differentiation proves (6). Every surviving character is quadratic, hence every coefficient is rational. If q clears their denominators, exponentiation yields

$$X_A^q = \prod_{\chi(R)=-1} u_\chi^{n_{A,\chi}} \qquad (n_{A,\chi}\in\mathbb{Z}), \tag{9}$$

after the exact factors in (5) have been absorbed into the integers $n_{A,\chi}$. This proves Proposition 3 and Theorem 1. □

The theorem is uniform; its proof does not enumerate the census. There are 1,560 nontrivial eligible occurrences in the frozen range, involving 2,232 supported character occurrences and 912 distinct quartic fields. These figures describe a verification queue, not a completeness claim here. Each application still certifies its primitive conductor, field, class numbers, Euler factors, exact norm map, primitive kernel, determinant, and orientation. The dimension-four and dimension-eight anchors independently replay (5), including relative index $I_\chi = 2$ in both dimension-eight cases.

2.2 Engine B: Shintani algebraicity and height rigidity

Engine B first verifies Shintani's unit-congruence and real-place hypotheses (the conditions labelled (0-3), (0-6), and (0-9) in [1]) and the index

$$[H : H \cap \mathbb{Q}^{\mathrm{ab}}] = 2. \tag{10}$$

The certificate prints the exact unit images modulo $\mathfrak{f}$, their signs at both real places, the splitting count, and the commutator subgroup; the word "verified" never means that these predicates were inferred from a group order alone.

Let $N/\mathbb{Q}$ be the actual normal closure and put $G_N = \mathrm{Gal}(N/\mathbb{Q})$. The fixed field of $[G_N, G_N]$ reconstructs the maximal absolutely abelian subfield. Independently, for each imaginary quadratic subfield k, the intrinsic ray group of k and its class-field polynomial reconstruct the same field. Equality is tested by exact field inclusions and conductor round-trips. The two routes must agree; this is not a numerical isomorphism heuristic.

Shintani's distribution relations introduce a finite transfer conductor $\mathfrak{c}$ over k. For every $\mathfrak{d} \mid \mathfrak{c}$, the proof record prints the ray class number, $w(\mathfrak{d})$, the distribution index $n(S_{\mathfrak{d}})$, and the conductor factor appearing in Shintani's theorem. Its prescribed multiple

$$m_{\mathfrak{d}} = 12\, c_{\mathfrak{d}}\, n(S_{\mathfrak{d}}) \tag{11}$$

clears that divisor term; in the trivial-divisor row $c_1 = h_k$. The safe exponent is the exact integer

$$m = \mathrm{lcm}_{\mathfrak{d}\mid\mathfrak{c}}\, m_{\mathfrak{d}}, \tag{12}$$

after also clearing the real-side distribution indices. The full table is part of each proof, rather than only its least common multiple. Shintani's theorem then places X_A^m in the certified ray field as an algebraic unit with the asserted Artin action.

The algebraic candidate is not obtained by trusting a decimal recognition. An exact relative polynomial for the ray field is certified over K by `rnfconductor`; its unit group is certified, and the reciprocal candidate polynomial is printed. Sturm sequences isolate its positive real roots. Exact Frobenius actions at two split primes label the roots by ray classes, producing an explicit candidate α_A for every A. Numerical recognition may have suggested this polynomial, but the proof begins only after these exact checks.

It remains to prove $X_A = \alpha_A$. The cone evaluator encloses every conjugate logarithmic difference by δ. Since Shintani's theorem makes both X_A^m and α_A^m algebraic,

$$\eta_A = (X_A/\alpha_A)^m$$

is an algebraic unit and the product formula gives

$$h(\eta_A) \le m\delta. \tag{13}$$

For degree $d \ge 3$, Voutier's explicit bound is

$$V(d) = \frac{1}{4d}\left(\frac{\log\log d}{\log d}\right)^3. \tag{14}$$

The certificate takes the minimum over every possible degree in its window. If the margin (2) exceeds 100, (13)–(14) force η_A to be a root of unity. Degree one is handled by the fact that a positive rational unit is one; degree two uses $\frac{1}{2}\log\phi$ as the non-torsion lower bound. Finally $\eta_A > 0$, so the root of unity is one, and $X_A = \alpha_A$.

Effective but not uniformly cheap. The required raw radius is $O(V_{\min}/m)$. Arb precision therefore grows with $\log m$, but conductor size, packet degree, and cone count also grow, and no uniform complexity claim is made. The order-ten case below has $m = 15840$ and remains comfortable; cases with much larger exponents can exceed the fixed per-case resource cap and belong to the frontier. This is a stated limit of Engine B, not evidence against its completed cases.

2.3 Engine C: CM descent and Stark's proved theorem

Let θ be one of the quartic characters in the support over K, and let $N/\mathbb{Q}$ be the splitting closure of its character field. Engine C is available when the projective quotient contains the required V_4, whose two imaginary quadratic fixed fields are denoted k_1, k_2. For either $k = k_i$, exact class-field reconstruction produces an abelian character field E/k and a finite set $\mathcal{C}$ of compatible linear ray characters ψ.

The phrase *linear reinduction* means the following exact statement, rather than projective equivalence:

$$\mathrm{Ind}_{G_K}^{G_\mathbb{Q}} \theta \cong \mathrm{Ind}_{G_k}^{G_\mathbb{Q}} \psi \tag{15}$$

as ordinary representations after restriction to the explicitly computed finite quotient $\mathrm{Gal}(N/\mathbb{Q})$. The characters on the two sides are compared on every conjugacy class. Multiplying ψ by a scalar character would generally change (15), which is why "scalar twists are forbidden." Among the remaining finite candidate set, exact Euler coefficients are appended until

$$\psi \longmapsto (a_n(\psi))_{n \in B} \tag{16}$$

is injective. Matching the source coefficients selects one ψ; the printed finite table proves uniqueness and replaces any informal appeal to numerical recognition.

Let E/k be the selected abelian extension of an imaginary quadratic field. Stark's proved rank-one theorem applies with a set S containing the complex place and the conductor primes. In every use of the global-unit conclusion below, $|S| \geq 3$ is checked explicitly, and $e = |\mu(E)|$ is computed exactly. If $\ell_g = \log |g\varepsilon|_{\mathrm{ord}}$, the normalization is

$$\zeta_S'(0, g) = -\frac{2}{e}\ell_g, \qquad \ell_g = -\frac{e}{2}\zeta_S'(0, g). \tag{17}$$

For the frozen quartic Fourier convention,

$$L_S'(0, \psi) = -\frac{4}{e}(\ell_1 - i\ell_\sigma), \qquad \ell_1 - i\ell_\sigma = -\frac{e}{4}L_S'(0, \psi). \tag{18}$$

These are formulas, not interchangeable shorthand. The $e = 2$ anchor cannot distinguish $2/e$ from $e/2$; the nontrivial normalization check is the $e = 4$ cross-route identity for RQ-000458. The specializations of (17)–(18) carry the same minus signs for $e = 6, 8, 12$.

For completeness, here is the analytic-to-algebraic step. The selected Hecke L'-value is enclosed independently by a theta/Mellin evaluation. Exact unit computation gives $E^\times/\mu(E)$, the Artin generator σ, and the anti-unit lattice

$$\ker(1 + \sigma^2) \subset E^\times/\mu(E).$$

Equations (17)–(18) turn the L'-ball into two logarithmic coordinates. Arb inversion of the resulting 2×2 real logarithm matrix must isolate one integral coordinate orbit; overlap with two integer orbits is a halt. Each isolated unit is then embedded into the common normal closure and multiplied by its exact complex conjugate. These positive CM norms are fixed by conjugation, generate the asserted one-place ray field over K, and are labeled by the exact Artin action. This proves the corresponding X_A, or its reciprocal, and the reciprocal packet polynomials below make that harmless. Roots of unity disappear from the CM norm, so the theorem claims the torsion-invariant packet rather than a literal representative of Stark's unit.

This gives a written proof scheme with finite hypotheses: exact reinduction (15), unique selection (16), Stark's proved formula (17)–(18), unique integral lattice isolation, and exact CM-norm identities. The machine-readable record supplies the large integer fields and coefficient tables but does not replace any logical step.

3 The order-six and order-ten packets

3.1 $\mathbb{Q}(\sqrt{7})$ at the ramified prime above seven

For $K = \mathbb{Q}(\sqrt{7})$ and $\mathfrak{m} = \mathfrak{p}_7\infty_2$, the one-place and two-place ray groups are C_6 and $C_6 \times C_2$. The support orders are two and six and (10) holds. The commutator-fixed calculation produces $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-7})$, and their intrinsic conductor reconstructions give the same degree-24 normal closure. The $\mathbb{Q}(i)$ route has two divisor rows and safe exponent 4032.

The packet polynomial over K is

$$\begin{aligned} P_7(X) = X^6 &- (6+2\sqrt{7})X^5 + (22+8\sqrt{7})X^4 \\ &- (34+13\sqrt{7})X^3 + (22+8\sqrt{7})X^2 \\ &- (6+2\sqrt{7})X + 1. \end{aligned} \tag{19}$$

It is irreducible over K, defines the one-place ray field, has six positive Sturm-isolated roots, and is Artin-labeled by the split primes 19 and 31.

For the class represented by $(3)^r$, an exact Yamamoto domain has generators

$$\lambda_r = (7-2\sqrt{7})/3^r, \qquad \lambda_r(8+3\sqrt{7}).$$

Its determinant is nine and its half-open parallelotope contains exactly nine lattice points. The resulting 27 independent double-sine evaluations are enclosed without a PARI L-value. At exponent 4032, the powered height upper bound is 9.1903×10^{-9}. The minimum Voutier lower bound in degrees three through 24 is 5.2279×10^{-5}, so the margin exceeds 5688. The degree-one and degree-two fallbacks finish the proof of (19).

This is the headline order-six instance. The same character order is the difficult component at the separate $\mathbb{Q}(\sqrt{21})$ wall, but here the index-two hypothesis holds and the packet closes. Thus the character order itself is not the obstruction.

3.2 Replications and the ramified-prime-3 control

For $\mathbb{Q}(\sqrt{14})$ at $\mathfrak{p}_7$, both $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-2})$ reconstruct the degree-24 normal closure. The latter route has clearing exponents 576 and 84, with least common multiple 4032. The packet is

$$\begin{aligned} P_{14}(X) = X^6 &- (13+4\sqrt{14})X^5 + (85+22\sqrt{14})X^4 \\ &- (139+38\sqrt{14})X^3 + (85+22\sqrt{14})X^2 \\ &- (13+4\sqrt{14})X + 1. \end{aligned} \tag{20}$$

Split primes 11 and 103 label the Artin generators. The 20-point cone evaluator gives powered height at most 7.1466×10^{-9}, hence margin greater than 7315.

The strongest boundary control is RQ-002057 over $\mathbb{Q}(\sqrt{57})$, with modulus of norm 27. The conductor is a power of the ramified prime 3 and the support still has order six. The bases $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-19})$ independently give the same degree-24 closure. On the shorter route the divisor exponents are $864, 324, 108$, hence $m = 2592$. The fundamental positive unit has order three modulo the finite modulus, so a ray-class domain is the union of three adjacent cones, with

240 exact affine points per class. The powered height is at most 6.982×10^{-8}, giving margin greater than 748. Thus order-six support and ramified-prime-3 structure are both reachable; neither alone explains the separate $\mathbb{Q}(\sqrt{21})$ boundary.

3.3 Three further Shintani closures

The three remaining rows of Theorem 2 use exactly the Engine-B proof above; we print their routes and algebraic candidates so that no theorem row is supported only by a corpus identifier.

For RQ-000108, $K = \mathbb{Q}(\sqrt{5})$ and $\mathrm{N}\,\mathfrak{f} = 45$. The two imaginary bases are $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-15})$, and both reconstruct the same degree-16 normal closure. With $y = (1+\sqrt{5})/2$, the relative packet polynomial is

$$X^4 - (1+6y)X^3 + (9+9y)X^2 - (1+6y)X + 1. \tag{21}$$

It is the certified one-place ray field over K. At $m = 2880$, the powered height is at most 2.12465×10^{-8}, so the Voutier margin exceeds 2460.

For RQ-000021, $K = \mathbb{Q}(\sqrt{2})$ and $\mathfrak{f} = (7)$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-14})$ reconstruct the same degree-24 closure. The packet polynomial is

$$\begin{aligned} X^6 &- (10+6\sqrt{2})X^5 + (58+40\sqrt{2})X^4 \\ &- (129+90\sqrt{2})X^3 + (58+40\sqrt{2})X^2 \\ &- (10+6\sqrt{2})X + 1. \end{aligned} \tag{22}$$

The cone boundary meets a split-prime face; the canonical value is the average of the upper and lower half-open conventions. Both are enclosed independently. At $m = 2016$, the powered height is at most 1.22671×10^{-8}, giving margin greater than 4261.

For RQ-002955, $K = \mathbb{Q}(\sqrt{77})$ and $\mathfrak{f} = \mathfrak{p}_7$ has HNF $\left[\begin{smallmatrix} 7 & 3 \\ 0 & 1 \end{smallmatrix}\right]$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-11})$ give the same degree-24 closure. Put $y = (1+\sqrt{77})/2$, so $y^2 - y - 19 = 0$. The packet is

$$\begin{aligned} X^6 &- (15+3y)X^5 + (107+26y)X^4 \\ &- (217+54y)X^3 + (107+26y)X^2 - (15+3y)X + 1. \end{aligned} \tag{23}$$

At $m = 4032$, the powered height is at most 1.01485×10^{-8}, and the margin exceeds 5151. In all three cases the exact ray-field identification, Sturm isolation, Artin labels, and the degree-one and degree-two fallbacks are included in the cited case records.

3.4 Order ten

For RQ-001107 over $\mathbb{Q}(\sqrt{33})$ and $\mathfrak{p}_{11}$, put $y = (1+\sqrt{33})/2 > 0$, so $y^2 - y - 8 = 0$. The packet is

$$\begin{aligned} P_{33}(X) = X^{10} &- (8+4y)X^9 + (74+30y)X^8 - (294+125y)X^7 \\ &+ (669+281y)X^6 - (871+368y)X^5 + (669+281y)X^4 \\ &- (294+125y)X^3 + (74+30y)X^2 - (8+4y)X + 1. \end{aligned} \tag{24}$$

Its absolute degree-20 field is the certified ray field over K. The actual packet-comparison degree is 40; the certificate conservatively covers degrees three through 80. On this whole window the minimum Voutier bound is

$$5.22795332222684\ldots \times 10^{-5}.$$

At $m = 15840$, the factor-100 raw-error target is $3.30047558221391\ldots \times 10^{-11}$; the achieved bound is 5.6731×10^{-13}, so the certified margin exceeds 5817. The degree-one and degree-two fallbacks are recorded separately. This support order is the first one in the present proved inventory divisible by the prime 5; its historical priority status is the survey-bounded status stated after Theorem 2.

4 CM-descent packets

4.1 A generic mixed-signature packet

The first packet completed by the generic Engine-C implementation is over $\mathbb{Q}(\sqrt{35})$, at the primitive ideal of norm 32. Both $\mathbb{Q}(\sqrt{-10})$ and $\mathbb{Q}(\sqrt{-14})$ have $e = 2$ and $|S| = 3$, but their class numbers are respectively 2 and 4. The real base itself has $h_K > 1$. Thus this is not a class-number-one coincidence: two intrinsically different CM reconstructions give the same Artin-labeled polynomial

$$\begin{aligned} P_{35}(X) = X^8 &- 38904X^7 + 905404X^6 + 62136X^5 \\ &- 873210X^4 + 62136X^3 + 905404X^2 - 38904X + 1. \end{aligned} \tag{25}$$

Exact conjugation fixes the four Artin norm classes, and Arb orbit isolation matches them to the four real roots. The remaining roots form two nonreal conjugate pairs, so the signature is $[4, 2]$. An exact conductor transport proves the norm-64 member.

The first $e = 6$ tranche comprises RQ-001569, RQ-007519, and RQ-001894. Six independent imaginary-base routes cover fourteen transported occurrences. Every route has $|S| \geq 3$, an exact character-selection signature, a certified theta/Mellin target, a unique integral anti-unit orbit, and an exact Artin bridge. The two routes for each canonical case give the identical degree-eight packet. For example, RQ-001569 is

$$\begin{aligned} X^8 &- 69087392X^7 + 7865869852X^6 - 660330848X^5 \\ &- 7321345850X^4 - 660330848X^3 + 7865869852X^2 - 69087392X + 1. \end{aligned} \tag{26}$$

The other two complete integer polynomials are printed in the reproducibility table's certificate.

4.2 $\mathbb{Q}(\sqrt{6})$ and auxiliary-prime independence

For RQ-000129 the two routes are over $\mathbb{Q}(\sqrt{-2})$, with $e = 8$, and $\mathbb{Q}(\sqrt{-3})$, with $e = 12$. The natural second route has only the complex place and one finite conductor prime, hence $|S| = 2$; the first attempt was halted because Stark's global-unit conclusion requires $|S| \geq 3$.

Let q be an auxiliary rational prime and write $P_q(Nq^{-s})$ for the removed local Euler factor. Since the primitive L-function has a simple zero at zero,

$$L'_{S\cup\{q\}}(0, \psi) = P_q(1)L'_S(0, \psi). \tag{27}$$

At $q = 3, 5$, exact Euler factors agree on the real source and both CM routes, and

$$P_3(1) = 1 + i, \qquad P_5(1) = 2. \tag{28}$$

Neither value vanishes, so the analytic rank remains one. On the secondary route the enlarged set has size three, and Stark's global-unit theorem now applies.

The exact Euler-factor identity (27), rather than a comparison of decimal approximations, controls the enlarged unit. Under (18), $q = 5$ doubles the exact logarithmic coordinates. Stark's uniqueness modulo $\mu(E)$ therefore says that the enlarged unit is the square of the natural unit modulo torsion; no arbitrary S-unit factor is available. The exact free unit lattice verifies this divisibility, and the positive CM norm kills torsion. Hence its unique positive square root recovers the natural norm. The $q = 3$ run independently gives the exact $(I \pm A)$ group-ring transform. Both enlarged computations were rerun from the Euler factor through lattice isolation; they were not copied from the halted $|S| = 2$ analytic attempt.

At the isolated level, the two routes have coordinates $(4, 0)$ and $(6, 0)$. Their positive norms q_8, q_{12} satisfy 256 exact common-normal-closure identities, summarized by

$$q_8^3 = q_{12}^2. \tag{29}$$

Dividing by 4 and 6 in the two exact free unit lattices gives primitive coordinates $(1, 0)$ on both sides. Their positive Artin norms agree. Indeed the source ray group is C_4, so (6) and (18), together with $\varepsilon_e = u_e^{e/2}$, give at each Artin label

$$\log X_A = -\log N_{E/E^+}(gu_e).$$

Thus X_A is the reciprocal of the primitive positive CM norm. The polynomial is reciprocal, so both have the identical polynomial

$$\boxed{X^8 - 8X^7 + 12X^6 + 8X^5 - 10X^4 + 8X^3 + 12X^2 - 8X + 1}. \tag{30}$$

An exact Sturm count gives four positive roots and no negative roots, as required by the selector-field signature $[4, 2]$.

The polynomial printed in the earlier draft,

$$X^8 + 24X^7 + 732X^6 - 9304X^5 + 53958X^4 - 9304X^3 + 732X^2 + 24X + 1,$$

has no real roots. It is the minimal polynomial of the complex anti-unit before division in the unit lattice and before taking the CM norm; it is not a packet polynomial. The correction record retains both objects and the exact maps between them.

4.3 Two disjoint theorem bases

For RQ-000458, $K = \mathbb{Q}(\sqrt{14})$, the finite ideal has HNF $\left[\begin{smallmatrix} 12 & 0 \\ 0 & 6 \end{smallmatrix}\right]$ and $\mathrm{Cl}_{\mathfrak{m}}(K) \simeq C_4 \times C_2$. Both routes identify

$$P_{458}(X) = X^4 - (20 + 6\sqrt{14})X^3 + (138 + 36\sqrt{14})X^2 - (20 + 6\sqrt{14})X + 1. \tag{31}$$

The Engine-B proof uses safe exponent 1152 and gives powered height 8.0791×10^{-9}, hence Voutier margin greater than 6470.

The independent Engine-C proof reinduces linearly over $\mathbb{Q}(\sqrt{-42})$ and $\mathbb{Q}(\sqrt{-3})$. On the first base, $|S| = 3$ and $e = 4$. The coefficient at $n = 7$ uniquely selects the ray character. Arb inversion isolates

$$(-8, 4), \quad (0, -4), \quad (8, -4), \quad (0, 4),$$

and 32 exact common-closure identities prove (31). Thus one theorem base is Shintani 1978 and the other is Stark 1980; they share no certificate intermediate. Because the second-route selection seal was not timestamped before execution, the process tag is `DUAL_ROUTED`, not the stronger `DUAL_PROVED`. This process caveat does not weaken either mathematical proof.

5 Structural boundary lemmas

Lemma 4 (No absolute-abelian fourth engine)**.** *A nontrivial one-place differenced invariant over a real quadratic field is not governed by an absolutely abelian ray field.*

Proof. If the one-place field $H/\mathbb{Q}$ were abelian, the complex conjugations above the two real places of K would be conjugate: any lift of the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$ interchanges the places. In an abelian group conjugate elements are equal. Every character of $\mathrm{Gal}(H/K)$, being a character of a subgroup of the finite abelian group $\mathrm{Gal}(H/\mathbb{Q})$, extends to an absolute character. Such a character has equal parity at the two real places. The Fourier support of (1), however, consists of characters odd at the retained place and even at the omitted place. Hence that support must be empty. □

Engine A is not an exception: it reduces individual quadratic characters by (2), but does not assert that the entire mixed-signature one-place ray field is absolutely abelian.

Lemma 5 (Index parity)**.** *Let H/K be a one-place ray field and let $N/\mathbb{Q}$ be its genuine normal closure. If the sign class R is nontrivial, then*

$$2 \mid [H : H \cap \mathbb{Q}^{\mathrm{ab}}]. \tag{32}$$

Equivalently, an odd Shintani index can occur only when the Fourier support of (1) is empty.

Proof. Put $G = \mathrm{Gal}(N/\mathbb{Q})$, $A = \mathrm{Gal}(N/K)$, and $B = \mathrm{Gal}(N/H)$. Let $c_1, c_2 \in A$ be the real-place inertia involutions. A lift of the nontrivial automorphism of $K/\mathbb{Q}$ interchanges them, so they are conjugate in G. Hence $c_1 c_2^{-1} \in [G, G]$. Since $G/A \simeq C_2$, one has $[G, G] \subseteq A$. Moreover B is normal in A, so $[G, G]B$ is a subgroup of A. In $A/B = \mathrm{Gal}(H/K)$, the involution at the omitted place is trivial and the one at the retained place is R. Thus the image of $[G, G]$ in A/B contains R. Its order is $[H : H \cap \mathbb{Q}^{\mathrm{ab}}]$, which is even if $R \neq 1$. Finally, characters of a finite abelian group separate a nonidentity element of order two, so $R = 1$ is equivalent to empty differenced support. □

The use of the genuine normal closure in Lemma 5 is essential. Conjugation coordinates at an unstable modulus do not define $[G, G]$. As an independent consistency check, a pre-registered replay over all 8,200 genuine normal-closure reconstructions found 446 odd indices greater than one. Every one had trivial sign class and empty differenced Fourier support; there were no exceptions. An earlier 88-row proxy population is excluded from this count and retained only as a provenance-control artifact.

6 Reproducibility and containment

The exact field computations use PARI/GP 2.15.4; analytic certificates use Python 3.12.3 and python-flint 0.9.0. The principal replays were run under Linux/x86-64 on an AMD EPYC 9354P virtual CPU. All analytic conclusions are Arb balls, never point estimates.

The principal machine-readable records are

result	record
first order six	`data/q7-p7-case-v1.json`
second order six	`data/q14-p7-case-v1.json`
RQ-000108	`data/rq000108-case-v1.json`
RQ-000021	`data/rq000021-case-v1.json`
ramified-prime-3 control	`data/q57-norm27-case-v1.json`
RQ-002955	`data/rq002955-case-v1.json`
order ten	`data/q33-p11-order10-case-v1.json`
generic CM packet	`artifacts/engine-c-w3-tranche-01-verified-v1.json`
$\mathbb{Q}(\sqrt{35})$ class numbers	`artifacts/q35-base-class-numbers-v1.json`
$e = 6$ tranche	`artifacts/engine-c-e6-tranche-01-verified-v1.json`
$\mathbb{Q}(\sqrt{6})$	`data/q6-norm8-case-v3.json`
two-route packet	`data/rq000458-dual-case-v1.json`
quadratic theorem	`data/engine-a-uniform-theorem-v1.json`
parity audit	`artifacts/results-paper-odd-index-parity-audit-v1.json`
seal-order audit	`artifacts/results-paper-seal-order-audit-v1.json`

Each record carries hashes of its exact and analytic dependencies. Failed attempts remain versioned. Predicate provenance is marked `GENUINE` or `PROXY`; no theorem tag may depend on a proxy. A later audit found an unmarked conjugation proxy in census classification, but none in a promoted theorem chain. The present paper was therefore frozen independently of all census populations and trends. This is a claim-boundary statement, not evidence for census completeness.

Seal ordering. The cycle-060 checkpoint was committed as `d1d355a14a` at 06:04:41 UTC. The generic Engine-C W3 tranche was sealed as `a0674aed11` at 07:07:17 UTC, and the first $e = 6$ tranche together with the original $\mathbb{Q}(\sqrt{6})$ closure was sealed as `0afdc1304d` at 08:41:02 UTC. All three precede the first manuscript commit. The later correction of the $\mathbb{Q}(\sqrt{6})$ polynomial illustrates the distinction between a properly sealed computation and a correct mathematical interpretation: seal order passed, but the positive-packet audit still found and repaired a semantic error.

Data availability. The companion archive has reserved DOI `10.5281/zenodo.21703306`. At the date of this referee draft the Zenodo record is an unpublished draft; publication of the archive is a hard gate before submission of the paper. The core file list and full SHA-256 values are in `artifacts/results-paper-core-manifest-v1.json`, whose SHA-256 is

`04ac746fe210d8b501345b242fadc165 fa83c7482b884178114bf6bd6e416f4c`.

Thus the paper names one content hash even before the DOI resolves; the released archive will include the complete repository subset, the failed-run transcripts, and one-command replay instructions.

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